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#include <iostream>
#include <stack>
using namespace std;

int main(){
    int currSet = 1; //Counter indicating which set we are on
    int numNames; //Variable for number of names in the set
    while (cin >> numNames && numNames != 0){
        stack<string> namesOdd; //If names !=0, then we have a set of names called
namesEven where we will store the names with an even index.
        cout << "SET " << currSet << endl; //Print which set we are on
        currSet = currSet + 1; //Update the set counter

        for (int i = 0; i < numNames; i++){ //loop to loop through the set of names
            string name;
            cin >> name; //take in the names
            if (i % 2 == 0){ //for the names, if the index % 2 == 0 (aka even)
                cout << name << endl; //directly print the name
            } else {
                namesOdd.push(name); //put the odd index names in the stack
            }
        }
        while (!namesOdd.empty()){ //When the stack has names in it
            cout << namesOdd.top() << endl; //Print the top value
            namesOdd.pop(); // remove that top value, and the while loop repeats til it
prints all of the values and no elems remain
        }
    }
    return 0;
}

```

//since the original list is in nondecending order, we can just print the even index values in the list as is, and then putting the odd index values in the stack in order, we can ensure that we print them in reverse order and in a way that ends in the largest names in the middle since the last elem added to the stack will be the largest name, getting smaller and smaller as the stack is emptied.